Numerical Analysis homework 3

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Abstract

My homework for numerical analysis.

Problem I: Cubic Spline Analysis

Problem Statement

Consider $s \in S_3^2$ on $[0, 2]$:

$$s(x) = \begin{cases} p(x) & \text{if } x \in [0, 1], \\ (2-x)^3 & \text{if } x \in [1, 2]. \end{cases}$$

Determine $p \in \mathbb{P}_3$ such that $s(0) = 0$. Is $s(x)$ a natural cubic spline?

Solution

Polynomial form and boundary condition: Let $p(x) = ax^3 + bx^2 + cx + d$. Using the boundary condition:

$$s(0) = p(0) = d = 0 \quad \Rightarrow \quad d = 0$$

Thus, $p(x) = ax^3 + bx^2 + cx$.

Continuity conditions at $x = 1$: Let $q(x) = (2-x)^3$. We require:

- **Function value continuity:**

$$p(1) = q(1) \Rightarrow a + b + c = (2-1)^3 = 1 \quad (1)$$

- **First derivative continuity:**

$$p'(x) = 3ax^2 + 2bx + c, \quad q'(x) = -3(2-x)^2$$

$$p'(1) = q'(1) \Rightarrow 3a + 2b + c = -3(2-1)^2 = -3 \quad (2)$$

- **Second derivative continuity:**

$$p''(x) = 6ax + 2b, \quad q''(x) = 6(2-x)$$

$$p''(1) = q''(1) \Rightarrow 6a + 2b = 6(2-1) = 6 \quad (3)$$

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Solving the system: We solve the system of equations:
From equation (3): $6a + 2b = 6 \Rightarrow 3a + b = 3 \quad$ (3a)
Subtract equation (1) from equation (2):

$$(3a + 2b + c) - (a + b + c) = -3 - 1 \Rightarrow 2a + b = -4 \quad (4)$$

Subtract equation (4) from equation (3a):

$$(3a + b) - (2a + b) = 3 - (-4) \Rightarrow a = 7$$

Substitute $a = 7$ into equation (3a):

$$3(7) + b = 3 \Rightarrow 21 + b = 3 \Rightarrow b = -18$$

Substitute $a = 7, b = -18$ into equation (1):

$$7 - 18 + c = 1 \Rightarrow -11 + c = 1 \Rightarrow c = 12$$

Compute:

$$s''(0) = p''(0) = 6a(0) + 2b = 2(-18) = -36 \neq 0$$
$$s''(2) = q''(2) = 6(2 - 2) = 0$$

Since $s''(0) \neq 0$, the spline is not natural.

Final Answers:

1. $p(x) = 7x^3 - 18x^2 + 12x$
2. $s(x)$ is **not** a natural cubic spline because $s''(0) = -36 \neq 0$

Problem II: Quadratic Spline Interpolation

Problem Statement

Given $f_i = f(x_i)$ of some scalar function at points $a = x_1 < x_2 < \cdots < x_n = b$, we consider interpolating f on $[a, b]$ with a quadratic spline $s \in \mathbb{S}_2^1$.

(a) Why is an additional condition needed to determine s uniquely?

(b) Define $m_i = s'(x_i)$ and $p_i = s|_{[x_i, x_{i+1}]}$. Determine p_i in terms of f_i, f_{i+1}, and m_i for $i = 1, 2, \ldots, n - 1$.

(c) Suppose $m_1 = f'(a)$ is given. Show how $m_2, m_3, \ldots, m_{n-1}$ can be computed.

Solution

(a) Uniqueness and additional condition: A quadratic spline $s \in \mathbb{S}_2^1$ on n nodes consists of $n-1$ quadratic polynomial pieces. Let's count the degrees of freedom and constraints precisely.

Degrees of freedom:

- Each quadratic polynomial piece has 3 parameters.
- Total parameters: $3(n - 1) = 3n - 3$.

Constraints:

- **Interpolation conditions:**
  - At the first node x_1: 1 condition ($s_1(x_1) = f_1$)
  - At the last node x_n: 1 condition ($s_{n-1}(x_n) = f_n$)

    * At each interior node x_i ($i = 2, \ldots, n-1$): 2 conditions ($s_{i-1}(x_i) = f_i$ and $s_i(x_i) = f_i$)
    * Total interpolation conditions: $1 + 1 + 2(n-2) = 2n - 2$

- **First derivative continuity conditions ($\mathbf{C^1}$):**
    * At each interior node x_i ($i = 2, \ldots, n-1$): 1 condition ($s'_{i-1}(x_i) = s'_i(x_i)$)
    * Total C^1 conditions: $n - 2$

Total constraints: $(2n-2) + (n-2) = 3n - 4$

Since we have $3n-3$ degrees of freedom but only $3n-4$ constraints, we are short by **one** condition to determine the spline uniquely.

(b) Expression for the quadratic polynomial piece: Using the given conditions:

- $p_i(x_i) = f_i$
- $p_i(x_{i+1}) = f_{i+1}$
- $p'_i(x_i) = m_i$

Let $p_i(x) = a_i(x - x_i)^2 + b_i(x - x_i) + c_i$. From the conditions:

- $p_i(x_i) = c_i = f_i$
- $p'_i(x_i) = b_i = m_i$
- $p_i(x_{i+1}) = a_i h_i^2 + m_i h_i + f_i = f_{i+1} \Rightarrow a_i = \dfrac{f_{i+1} - f_i - m_i h_i}{h_i^2}$

Therefore, the quadratic polynomial is:

$$p_i(x) = \frac{f_{i+1} - f_i - m_i h_i}{h_i^2}(x - x_i)^2 + m_i(x - x_i) + f_i$$

where $h_i = x_{i+1} - x_i$.

(c) Recursive computation of m_i: Using the expression for $p_i(x)$ from part (b):

$$p_i(x) = \frac{f_{i+1} - f_i - m_i h_i}{h_i^2}(x - x_i)^2 + m_i(x - x_i) + f_i$$

Differentiate to find the derivative:

$$p'_i(x) = \frac{2(f_{i+1} - f_i - m_i h_i)}{h_i^2}(x - x_i) + m_i$$

Now evaluate at the right endpoint $x = x_{i+1}$:

$$p'_i(x_{i+1}) = \frac{2(f_{i+1} - f_i - m_i h_i)}{h_i^2} \cdot h_i + m_i = \frac{2(f_{i+1} - f_i - m_i h_i)}{h_i} + m_i$$

Simplify:

$$p'_i(x_{i+1}) = \frac{2(f_{i+1} - f_i)}{h_i} - 2m_i + m_i = \frac{2(f_{i+1} - f_i)}{h_i} - m_i$$

By the continuity of the first derivative (C^1 continuity) at interior nodes, we have:

$$p'_i(x_{i+1}) = p'_{i+1}(x_{i+1}) = m_{i+1}$$

Therefore, we obtain the recurrence relation:

$$m_{i+1} = \frac{2(f_{i+1} - f_i)}{h_i} - m_i, \quad \text{for } i = 1, 2, \ldots, n-1$$

Given $m_1 = f'(a)$, we can compute sequentially:

$$\begin{aligned} m_2 &= \frac{2(f_2 - f_1)}{h_1} - m_1 \\ m_3 &= \frac{2(f_3 - f_2)}{h_2} - m_2 \\ &\vdots \\ m_{n-1} &= \frac{2(f_{n-1} - f_{n-2})}{h_{n-2}} - m_{n-2} \end{aligned}$$

This recurrence allows us to compute all the derivatives $m_2, m_3, \ldots, m_n$ starting from the given initial condition m_1.

Problem III: Natural Cubic Spline Construction

Problem Statement

Let $s_1(x) = 1 + c(x+1)^3$ where $x \in [-1, 0]$ and $c \in \mathbb{R}$. Determine $s_2(x)$ on $[0, 1]$ such that

$$s(x) = \begin{cases} s_1(x) & \text{if } x \in [-1, 0], \\ s_2(x) & \text{if } x \in [0, 1] \end{cases}$$

is a natural cubic spline on $[-1, 1]$ with knots $-1, 0, 1$. How must c be chosen if one wants $s(1) = -1$?

Solution

Step 1: General form of $s_2(x)$ Since s is a cubic spline, $s_2(x)$ must be a cubic polynomial on $[0, 1]$. Let

$$s_2(x) = A + Bx + Cx^2 + Dx^3.$$

Step 2: Continuity conditions at $x = 0$ We require C^2 continuity at the knot $x = 0$.
Function value continuity:

$$s_1(0) = s_2(0) \Rightarrow 1 + c(0+1)^3 = A \Rightarrow 1 + c = A. \quad (1)$$

First derivative continuity:

$$s_1'(x) = 3c(x+1)^2, \quad s_2'(x) = B + 2Cx + 3Dx^2.$$
$$s_1'(0) = s_2'(0) \Rightarrow 3c(1)^2 = B \Rightarrow 3c = B. \quad (2)$$

Second derivative continuity:

$$s_1''(x) = 6c(x+1), \quad s_2''(x) = 2C + 6Dx.$$
$$s_1''(0) = s_2''(0) \Rightarrow 6c(1) = 2C \Rightarrow 6c = 2C \Rightarrow C = 3c. \quad (3)$$

Step 3: Natural boundary conditions A natural cubic spline requires zero second derivatives at the endpoints.

- At $x = -1$:
$$s_1''(-1) = 6c(-1+1) = 0$$
- At $x = 1$:
$$s_2''(1) = 2C + 6D(1) = 0 \Rightarrow 2C + 6D = 0. \quad (4)$$

Step 4: Solve for coefficients From (3): $C = 3c$. Substitute into (4):

$$2(3c) + 6D = 0 \Rightarrow 6c + 6D = 0 \Rightarrow D = -c.$$

So far we have:

$$A = 1 + c, \quad B = 3c, \quad C = 3c, \quad D = -c.$$

Thus:

$$s_2(x) = (1 + c) + 3cx + 3cx^2 - cx^3.$$

Step 5: Additional condition $s(1) = -1$

$$s_2(1) = (1 + c) + 3c(1) + 3c(1)^2 - c(1)^3 = 1 + c + 3c + 3c - c = 1 + 6c.$$

Set $s_2(1) = -1$:

$$1 + 6c = -1 \Rightarrow 6c = -2 \Rightarrow c = -\frac{1}{3}.$$

Step 6: Final expression for $s_2(x)$ Substitute $c = -\frac{1}{3}$ into $s_2(x)$:

$$s_2(x) = \left(1 - \frac{1}{3}\right) + 3\left(-\frac{1}{3}\right)x + 3\left(-\frac{1}{3}\right)x^2 - \left(-\frac{1}{3}\right)x^3$$

$$s_2(x) = \frac{2}{3} - x - x^2 + \frac{1}{3}x^3.$$

Final Answers:

1. The cubic polynomial on $[0, 1]$ is:

$$s_2(x) = (1 + c) + 3cx + 3cx^2 - cx^3$$

2. For $s(1) = -1$, we must choose:

$$c = -\frac{1}{3}$$

and then

$$s_2(x) = \frac{2}{3} - x - x^2 + \frac{1}{3}x^3$$

Problem IV: Natural Cubic Spline and Bending Energy

Problem Statement

Consider $f(x) = \cos\left(\frac{\pi}{2}x\right)$ with $x \in [-1, 1]$.

(a) Determine the natural cubic spline interpolant to f on knots $-1, 0, 1$.

(b) As discussed in the class, natural cubic splines have the minimal total bending energy. Verify this by taking $g(x)$ to be (i) the quadratic polynomial that interpolates f at $-1, 0, 1$, and (ii) $f(x)$.

Solution

(a) Natural cubic spline interpolant Let the cubic spline be defined as:

$$s(x) = \begin{cases} s_1(x) = a_1 + b_1(x + 1) + c_1(x + 1)^2 + d_1(x + 1)^3, & x \in [-1, 0] \\ s_2(x) = a_2 + b_2x + c_2x^2 + d_2x^3, & x \in [0, 1] \end{cases}$$

First, compute the function values at the knots:

$$f(-1) = \cos\left(-\frac{\pi}{2}\right) = 0, \quad f(0) = \cos(0) = 1, \quad f(1) = \cos\left(\frac{\pi}{2}\right) = 0$$

Interpolation conditions:

$$s_1(-1) = a_1 = 0 \quad (1)$$
$$s_1(0) = a_1 + b_1 + c_1 + d_1 = 1 \quad (2)$$
$$s_2(0) = a_2 = 1 \quad (3)$$
$$s_2(1) = a_2 + b_2 + c_2 + d_2 = 0 \quad (4)$$

Continuity of first derivative at $x = 0$:

$$s_1'(x) = b_1 + 2c_1(x+1) + 3d_1(x+1)^2$$
$$s_2'(x) = b_2 + 2c_2x + 3d_2x^2$$
$$s_1'(0) = b_1 + 2c_1 + 3d_1 = s_2'(0) = b_2 \quad (5)$$

Continuity of second derivative at $x = 0$:

$$s_1''(x) = 2c_1 + 6d_1(x+1)$$
$$s_2''(x) = 2c_2 + 6d_2x$$
$$s_1''(0) = 2c_1 + 6d_1 = s_2''(0) = 2c_2 \quad (6)$$

Natural boundary conditions:

$$s_1''(-1) = 2c_1 = 0 \Rightarrow c_1 = 0 \quad (7)$$
$$s_2''(1) = 2c_2 + 6d_2 = 0 \quad (8)$$

Solve the system: From (7): $c_1 = 0$
From (1): $a_1 = 0$
From (3): $a_2 = 1$
From (2): $b_1 + d_1 = 1 \quad$ (2a)
From (6): $6d_1 = 2c_2 \Rightarrow c_2 = 3d_1 \quad$ (6a)
From (8): $2c_2 + 6d_2 = 0 \Rightarrow 6d_1 + 6d_2 = 0 \Rightarrow d_2 = -d_1 \quad$ (8a)
From (4): $1 + b_2 + c_2 + d_2 = 0 \Rightarrow b_2 + c_2 + d_2 = -1 \quad$ (4a)
Substitute (6a) and (8a) into (4a):

$$b_2 + 3d_1 - d_1 = -1 \Rightarrow b_2 + 2d_1 = -1 \quad (4b)$$

From (5): $b_2 = b_1 + 3d_1 \quad$ (5a)
Substitute (5a) into (4b):

$$b_1 + 3d_1 + 2d_1 = -1 \Rightarrow b_1 + 5d_1 = -1 \quad (4c)$$

Now solve the system of (2a) and (4c):

$$\begin{cases} b_1 + d_1 = 1 \\ b_1 + 5d_1 = -1 \end{cases}$$

Subtract the first equation from the second:

$$4d_1 = -2 \Rightarrow d_1 = -\frac{1}{2}$$

Then:

$$b_1 = 1 - d_1 = 1 + \frac{1}{2} = \frac{3}{2}$$

From (5a): $b_2 = b_1 + 3d_1 = \frac{3}{2} - \frac{3}{2} = 0$
From (6a): $c_2 = 3d_1 = -\frac{3}{2}$
From (8a): $d_2 = -d_1 = \frac{1}{2}$

Final spline functions:

$$s_1(x) = \frac{3}{2}(x+1) - \frac{1}{2}(x+1)^3, \quad x \in [-1, 0]$$
$$s_2(x) = 1 - \frac{3}{2}x^2 + \frac{1}{2}x^3, \quad x \in [0, 1]$$

(b) Bending energy comparison The bending energy is defined as:

$$E[g] = \int_{-1}^{1} [g''(x)]^2 dx$$

(i) Quadratic polynomial interpolant: Let $g(x) = Ax^2 + Bx + C$. Using interpolation conditions:

$$\begin{aligned} g(-1) &= A - B + C = 0 \\ g(0) &= C = 1 \\ g(1) &= A + B + C = 0 \end{aligned}$$

Solving: $C = 1$, and from the first and third equations:

$$A - B = -1, \quad A + B = -1$$

Adding: $2A = -2 \Rightarrow A = -1$, then $B = 0$.
So $g(x) = -x^2 + 1$, and $g''(x) = -2$.
Bending energy:

$$E[g] = \int_{-1}^{1} (-2)^2 dx = 4 \times 2 = 8$$

(ii) Original function $f(x)$:

$$f(x) = \cos\left(\frac{\pi}{2}x\right), \quad f'(x) = -\frac{\pi}{2}\sin\left(\frac{\pi}{2}x\right), \quad f''(x) = -\frac{\pi^2}{4}\cos\left(\frac{\pi}{2}x\right)$$

Bending energy:

$$E[f] = \int_{-1}^{1} \left[-\frac{\pi^2}{4}\cos\left(\frac{\pi}{2}x\right)\right]^2 dx = \frac{\pi^4}{16}\int_{-1}^{1} \cos^2\left(\frac{\pi}{2}x\right) dx$$

Using the identity $\cos^2\theta = \frac{1+\cos 2\theta}{2}$:

$$E[f] = \frac{\pi^4}{16}\int_{-1}^{1} \frac{1+\cos(\pi x)}{2} dx = \frac{\pi^4}{32}\left[\int_{-1}^{1} 1 dx + \int_{-1}^{1} \cos(\pi x) dx\right]$$

Since $\int_{-1}^{1} \cos(\pi x) dx = \left[\frac{\sin(\pi x)}{\pi}\right]_{-1}^{1} = 0$, we have:

$$E[f] = \frac{\pi^4}{32} \times 2 = \frac{\pi^4}{16} \approx \frac{97.4091}{16} \approx 6.088$$

(iii) Natural cubic spline $s(x)$: Compute second derivatives:

$$s_1''(x) = -3(x+1), \quad s_2''(x) = -3 + 3x$$

Bending energy:

$$\begin{aligned} E[s] &= \int_{-1}^{0} [-3(x+1)]^2 dx + \int_{0}^{1} [-3+3x]^2 dx \\ &= 9\int_{-1}^{0} (x+1)^2 dx + 9\int_{0}^{1} (1-x)^2 dx \\ &= 9\left[\frac{(x+1)^3}{3}\right]_{-1}^{0} + 9\left[-\frac{(1-x)^3}{3}\right]_{0}^{1} \\ &= 3[(1)^3 - (0)^3] + 3[0 - (-1)^3] = 3(1) + 3(1) = 6 \end{aligned}$$

Comparison:

$$E[s] = 6 < E[f] \approx 6.088 < E[g] = 8$$

This verifies that the natural cubic spline has the minimal bending energy.

Problem V: The Quadratic B-spline $B_i^2(x)$

Problem Statement

(a) Derive the same explicit expression of $B_i^2(x)$ as that in the notes from the recursive definition of B-splines and the hat function.

(b) Verify that $\frac{d}{dx}B_i^2(x)$ is continuous at t_i and t_{i+1}.

(c) Show that only one $x^* \in (t_{i-1}, t_{i+2})$ satisfies $\frac{d}{dx}B_i^2(x^*) = 0$. Express x^* in terms of the knots within the interval of support.

(d) Consequently, show $B_i^2(x) \in [0, 1)$.

(e) Plot $B_i^2(x)$ for $t_i = i$.

Solution

(a) Derivation of explicit expression Using the recursive definition:

$$B_i^{n+1}(x) = \frac{x - t_{i-1}}{t_{i+n} - t_{i-1}} B_i^n(x) + \frac{t_{i+n+1} - x}{t_{i+n+1} - t_i} B_{i+1}^n(x)$$

with base case:

$$B_i^0(x) = \begin{cases} 1 & \text{if } x \in (t_{i-1}, t_i) \\ 0 & \text{otherwise} \end{cases}$$

For quadratic B-splines ($n = 2$), we have:

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} B_i^1(x) + \frac{t_{i+2} - x}{t_{i+2} - t_i} B_{i+1}^1(x)$$

First, compute the linear B-splines:

$$B_i^1(x) = \frac{x - t_{i-1}}{t_i - t_{i-1}} B_i^0(x) + \frac{t_{i+1} - x}{t_{i+1} - t_i} B_{i+1}^0(x)$$

$$B_{i+1}^1(x) = \frac{x - t_i}{t_{i+1} - t_i} B_{i+1}^0(x) + \frac{t_{i+2} - x}{t_{i+2} - t_{i+1}} B_{i+2}^0(x)$$

The support of $B_i^2(x)$ is (t_{i-1}, t_{i+2}). We analyze piece by piece:

Interval 1: (t_{i-1}, t_i) On this interval: $B_i^0(x) = 1$, $B_{i+1}^0(x) = 0$, $B_{i+2}^0(x) = 0$

Thus:

$$B_i^1(x) = \frac{x - t_{i-1}}{t_i - t_{i-1}}, \quad B_{i+1}^1(x) = 0$$

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} \cdot \frac{x - t_{i-1}}{t_i - t_{i-1}} = \frac{(x - t_{i-1})^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}$$

Interval 2: (t_i, t_{i+1}) On this interval: $B_i^0(x) = 0$, $B_{i+1}^0(x) = 1$, $B_{i+2}^0(x) = 0$

Thus:

$$B_i^1(x) = \frac{t_{i+1} - x}{t_{i+1} - t_i}, \quad B_{i+1}^1(x) = \frac{x - t_i}{t_{i+1} - t_i}$$

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} \cdot \frac{t_{i+1} - x}{t_{i+1} - t_i} + \frac{t_{i+2} - x}{t_{i+2} - t_i} \cdot \frac{x - t_i}{t_{i+1} - t_i}$$

Interval 3: (t_{i+1}, t_{i+2}) On this interval: $B_i^0(x) = 0$, $B_{i+1}^0(x) = 0$, $B_{i+2}^0(x) = 1$

Thus:

$$B_i^1(x) = 0, \quad B_{i+1}^1(x) = \frac{t_{i+2} - x}{t_{i+2} - t_{i+1}}$$

$$B_i^2(x) = \frac{t_{i+2} - x}{t_{i+2} - t_i} \cdot \frac{t_{i+2} - x}{t_{i+2} - t_{i+1}} = \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}$$

This matches the expression given in Example 3.25.

(b) Continuity of derivative We verify that $\frac{d}{dx} B_i^2(x)$ is continuous at both t_i and t_{i+1}.

At $x = t_i$: From Interval 1 (t_{i-1}, t_i):

$$B_i^2(x) = \frac{(x - t_{i-1})^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}$$

$$\frac{d}{dx} B_i^2(x) = \frac{2(x - t_{i-1})}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}$$

At $x = t_i^-$:

$$\frac{d}{dx} B_i^2(t_i^-) = \frac{2(t_i - t_{i-1})}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})} = \frac{2}{t_{i+1} - t_{i-1}}$$

From Interval 2 (t_i, t_{i+1}):

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} \cdot \frac{t_{i+1} - x}{t_{i+1} - t_i} + \frac{t_{i+2} - x}{t_{i+2} - t_i} \cdot \frac{x - t_i}{t_{i+1} - t_i}$$

Differentiate term by term:

$$\begin{aligned}\frac{d}{dx} B_i^2(x) =& \frac{1}{t_{i+1} - t_{i-1}} \cdot \frac{t_{i+1} - x}{t_{i+1} - t_i} + \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} \cdot \frac{-1}{t_{i+1} - t_i} \\ &+ \frac{-1}{t_{i+2} - t_i} \cdot \frac{x - t_i}{t_{i+1} - t_i} + \frac{t_{i+2} - x}{t_{i+2} - t_i} \cdot \frac{1}{t_{i+1} - t_i}\end{aligned}$$

At $x = t_i^+$:

$$\begin{aligned}\frac{d}{dx} B_i^2(t_i^+) =& \frac{1}{t_{i+1} - t_{i-1}} \cdot \frac{t_{i+1} - t_i}{t_{i+1} - t_i} + \frac{t_i - t_{i-1}}{t_{i+1} - t_{i-1}} \cdot \frac{-1}{t_{i+1} - t_i} \\ &+ \frac{-1}{t_{i+2} - t_i} \cdot \frac{t_i - t_i}{t_{i+1} - t_i} + \frac{t_{i+2} - t_i}{t_{i+2} - t_i} \cdot \frac{1}{t_{i+1} - t_i}\end{aligned}$$

Simplify:

$$\begin{aligned}\frac{d}{dx} B_i^2(t_i^+) =& \frac{1}{t_{i+1} - t_{i-1}} - \frac{t_i - t_{i-1}}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + 0 + \frac{1}{t_{i+1} - t_i} \\ =& \frac{1}{t_{i+1} - t_{i-1}} + \frac{1}{t_{i+1} - t_i} - \frac{t_i - t_{i-1}}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)}\end{aligned}$$

Combine over common denominator $(t_{i+1} - t_{i-1})(t_{i+1} - t_i)$:

$$\begin{aligned}\frac{d}{dx} B_i^2(t_i^+) =& \frac{t_{i+1} - t_i + t_{i+1} - t_{i-1} - (t_i - t_{i-1})}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} \\ =& \frac{t_{i+1} - t_i + t_{i+1} - t_{i-1} - t_i + t_{i-1}}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} \\ =& \frac{2(t_{i+1} - t_i)}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} = \frac{2}{t_{i+1} - t_{i-1}}\end{aligned}$$

Thus: $\frac{d}{dx} B_i^2(t_i^-) = \frac{2}{t_{i+1} - t_{i-1}} = \frac{d}{dx} B_i^2(t_i^+)$

At $x = t_{i+1}$: From Interval 2 (t_i, t_{i+1}): At $x = t_{i+1}^-$:

$$\begin{aligned}
\frac{d}{dx}B_i^2(t_{i+1}^-) =& \frac{t_{i+1} + t_{i-1} - 2t_{i+1}}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{t_{i+2} + t_i - 2t_{i+1}}{(t_{i+2} - t_i)(t_{i+1} - t_i)} \\
=& \frac{t_{i-1} - t_{i+1}}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{t_{i+2} + t_i - 2t_{i+1}}{(t_{i+2} - t_i)(t_{i+1} - t_i)} \\
=& \frac{-(t_{i+1} - t_{i-1})}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{t_{i+2} + t_i - 2t_{i+1}}{(t_{i+2} - t_i)(t_{i+1} - t_i)} \\
=& \frac{-1}{t_{i+1} - t_i} + \frac{t_{i+2} + t_i - 2t_{i+1}}{(t_{i+2} - t_i)(t_{i+1} - t_i)}
\end{aligned}$$

From Interval 3 (t_{i+1}, t_{i+2}):

$$B_i^2(x) = \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}$$

$$\frac{d}{dx}B_i^2(x) = \frac{-2(t_{i+2} - x)}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}$$

At $x = t_{i+1}^+$:

$$\frac{d}{dx}B_i^2(t_{i+1}^+) = \frac{-2(t_{i+2} - t_{i+1})}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})} = \frac{-2}{t_{i+2} - t_i}$$

Now show equality:

$$\begin{aligned}
\frac{d}{dx}B_i^2(t_{i+1}^-) =& \frac{-1}{t_{i+1} - t_i} + \frac{t_{i+2} + t_i - 2t_{i+1}}{(t_{i+2} - t_i)(t_{i+1} - t_i)} \\
=& \frac{-(t_{i+2} - t_i) + (t_{i+2} + t_i - 2t_{i+1})}{(t_{i+2} - t_i)(t_{i+1} - t_i)} \\
=& \frac{-t_{i+2} + t_i + t_{i+2} + t_i - 2t_{i+1}}{(t_{i+2} - t_i)(t_{i+1} - t_i)} \\
=& \frac{2t_i - 2t_{i+1}}{(t_{i+2} - t_i)(t_{i+1} - t_i)} = \frac{-2(t_{i+1} - t_i)}{(t_{i+2} - t_i)(t_{i+1} - t_i)} = \frac{-2}{t_{i+2} - t_i}
\end{aligned}$$

Thus: $\frac{d}{dx}B_i^2(t_{i+1}^-) = \frac{-2}{t_{i+2} - t_i} = \frac{d}{dx}B_i^2(t_{i+1}^+)$

(c) Unique critical point We analyze the derivative on each interval to show there exists exactly one critical point.

Interval 1: (t_{i-1}, t_i)

$$\frac{d}{dx}B_i^2(x) = \frac{2(x - t_{i-1})}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}$$

Since $x - t_{i-1} > 0$ and all denominators are positive, $\frac{d}{dx}B_i^2(x) > 0$ for all $x \in (t_{i-1}, t_i)$. No critical points in this interval.

Interval 3: (t_{i+1}, t_{i+2})

$$\frac{d}{dx}B_i^2(x) = \frac{-2(t_{i+2} - x)}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}$$

Since $t_{i+2} - x > 0$ and all denominators are positive, $\frac{d}{dx}B_i^2(x) < 0$ for all $x \in (t_{i+1}, t_{i+2})$. No critical points in this interval.

Interval 2: (t_i, t_{i+1}) The derivative is:

$$\frac{d}{dx}B_i^2(x) = \frac{t_{i+1} + t_{i-1} - 2x}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{t_{i+2} + t_i - 2x}{(t_{i+2} - t_i)(t_{i+1} - t_i)}$$

Set $\frac{d}{dx}B_i^2(x) = 0$:

$$\frac{t_{i+1} + t_{i-1} - 2x}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{t_{i+2} + t_i - 2x}{(t_{i+2} - t_i)(t_{i+1} - t_i)} = 0$$

Multiply through by $(t_{i+1} - t_i)$:

$$\frac{t_{i+1} + t_{i-1} - 2x}{t_{i+1} - t_{i-1}} + \frac{t_{i+2} + t_i - 2x}{t_{i+2} - t_i} = 0$$

Let $A = \frac{1}{t_{i+1}-t_{i-1}}$, $B = \frac{1}{t_{i+2}-t_i}$. Then:

$$A(t_{i+1} + t_{i-1} - 2x) + B(t_{i+2} + t_i - 2x) = 0$$

$$A(t_{i+1} + t_{i-1}) + B(t_{i+2} + t_i) - 2(A + B)x = 0$$

$$x^* = \frac{A(t_{i+1} + t_{i-1}) + B(t_{i+2} + t_i)}{2(A + B)}$$

Substitute back A and B:

$$x^* = \frac{\frac{t_{i+1}+t_{i-1}}{t_{i+1}-t_{i-1}} + \frac{t_{i+2}+t_i}{t_{i+2}-t_i}}{2\left(\frac{1}{t_{i+1}-t_{i-1}} + \frac{1}{t_{i+2}-t_i}\right)}$$

Since the derivative is strictly positive on (t_{i-1}, t_i), strictly negative on (t_{i+1}, t_{i+2}), and continuous on $[t_i, t_{i+1}]$, by the Intermediate Value Theorem there exists exactly one $x^* \in (t_i, t_{i+1})$ where $\frac{d}{dx}B_i^2(x^*) = 0$.

(d) Range of $B_i^2(x)$ The quadratic B-splines satisfy the **partition of unity** property:

$$\sum_j B_j^2(x) = 1 \quad \text{for all } x$$

Additionally, each quadratic B-spline is non-negative:

$$B_j^2(x) \geq 0 \quad \text{for all } j \text{ and all } x$$

Since the sum of all quadratic B-splines equals 1 and each is non-negative, no individual quadratic B-spline can attain the value 1, unless all other B-splines in the sum are zero at that point. However, due to the local support property of B-splines, for any given x, there are always at least two non-zero quadratic B-splines in the sum.

More precisely, for any x in the interior of the domain, there exist at least two indices j such that $B_j^2(x) > 0$. Therefore, for each individual quadratic B-spline:

$$B_i^2(x) \leq 1 - \sum_{j \neq i} B_j^2(x) < 1$$

Thus, we conclude that:

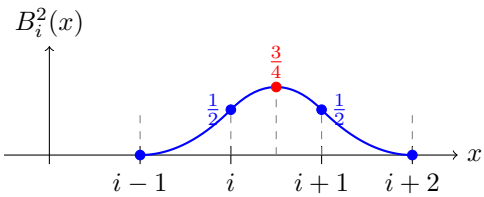
$$B_i^2(x) \in [0, 1) \quad \text{for all } x$$

This completes the proof that the quadratic B-spline takes values in the interval $[0, 1)$.

(e) Plot for $t_i = i$

$$B_i^2(x) = \begin{cases} \frac{(x-i+1)^2}{2} & \text{if } x \in (i-1, i]; \\ \frac{3}{4} - \left(x - (i + \frac{1}{2})\right)^2 & \text{if } x \in (i, i+1]; \\ \frac{(i+2-x)^2}{2} & \text{if } x \in (i+1, i+2]; \\ 0 & \text{otherwise.} \end{cases}$$

The plot of $B_i^2(x)$ is:



Quadratic B-spline $B_i^2(x)$ with uniform knots $t_j = j$

Problem VI: Verification of Theorem 3.32 for Quadratic Case

Problem Statement

Verify Theorem 3.32 algebraically for the case of $n = 2$, i.e.

$$(t_{i+2} - t_{i-1})[t_{i-1}, t_i, t_{i+1}, t_{i+2}](t - x)_+^2 = B_i^2(x)$$

Solution

The truncated power $(t-x)_+^2$ is non-zero only when $t > x$. Therefore, the value of the divided difference depends on the position of x relative to the knots.

Case 1: $x \geq t_{i+2}$ If $x \geq t_{i+2}$, then for all $t = t_{i-1}, t_i, t_{i+1}, t_{i+2}$, we have $t \leq x$, so $(t - x)_+^2 = 0$. Thus:

$$F(x) = 0 = B_i^2(x)$$

Case 2: $x \in [t_{i+1}, t_{i+2})$ If $x \in [t_{i+1}, t_{i+2})$, then:

- For $t = t_{i-1}, t_i, t_{i+1}$: $t \leq x$, so $(t - x)_+^2 = 0$
- For $t = t_{i+2}$: $t > x$, so $(t_{i+2} - x)_+^2 = (t_{i+2} - x)^2$

Compute the divided difference:

$$\begin{aligned}
f[t_{i+2}] &= (t_{i+2} - x)^2 \\
f[t_{i+1}, t_{i+2}] &= \frac{(t_{i+2} - x)^2 - 0}{t_{i+2} - t_{i+1}} = \frac{(t_{i+2} - x)^2}{t_{i+2} - t_{i+1}} \\
f[t_i, t_{i+1}, t_{i+2}] &= \frac{f[t_{i+1}, t_{i+2}] - f[t_i, t_{i+1}]}{t_{i+2} - t_i} = \frac{\frac{(t_{i+2}-x)^2}{t_{i+2}-t_{i+1}} - 0}{t_{i+2} - t_i} = \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})} \\
f[t_{i-1}, t_i, t_{i+1}, t_{i+2}] &= \frac{f[t_i, t_{i+1}, t_{i+2}] - f[t_{i-1}, t_i, t_{i+1}]}{t_{i+2} - t_{i-1}} = \frac{\frac{(t_{i+2}-x)^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})} - 0}{t_{i+2} - t_{i-1}} \\
&= \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_{i-1})(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}
\end{aligned}$$

Multiply by $(t_{i+2} - t_{i-1})$:

$$(t_{i+2} - t_{i-1})F(x) = \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}$$

This matches the third piece of $B_i^2(x)$ from Example 3.25.

Case 3: $x \in [t_i, t_{i+1})$ For $x \in [t_i, t_{i+1})$:

- For $t = t_{i-1}, t_i$: $t \leq x$, so $(t - x)_+^2 = 0$
- For $t = t_{i+1}, t_{i+2}$: $t > x$, so $(t - x)_+^2 = (t - x)^2$

Compute the divided difference step by step:

$$\begin{aligned}
f[t_{i-1}] &= 0 \\
f[t_i] &= 0 \\
f[t_{i+1}] &= (t_{i+1} - x)^2 \\
f[t_{i+2}] &= (t_{i+2} - x)^2
\end{aligned}$$

First order divided differences:

$$f[t_{i-1}, t_i] = \frac{0-0}{t_i - t_{i-1}} = 0$$
$$f[t_i, t_{i+1}] = \frac{(t_{i+1}-x)^2 - 0}{t_{i+1} - t_i} = \frac{(t_{i+1}-x)^2}{t_{i+1} - t_i}$$
$$f[t_{i+1}, t_{i+2}] = \frac{(t_{i+2}-x)^2 - (t_{i+1}-x)^2}{t_{i+2} - t_{i+1}} = \frac{(t_{i+2}-t_{i+1})(t_{i+2}+t_{i+1}-2x)}{t_{i+2}-t_{i+1}} = t_{i+2}+t_{i+1}-2x$$

Second order divided differences:

$$f[t_{i-1}, t_i, t_{i+1}] = \frac{f[t_i, t_{i+1}] - f[t_{i-1}, t_i]}{t_{i+1} - t_{i-1}} = \frac{\frac{(t_{i+1}-x)^2}{t_{i+1}-t_i} - 0}{t_{i+1} - t_{i-1}} = \frac{(t_{i+1}-x)^2}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)}$$
$$f[t_i, t_{i+1}, t_{i+2}] = \frac{f[t_{i+1}, t_{i+2}] - f[t_i, t_{i+1}]}{t_{i+2} - t_i} = \frac{(t_{i+2}+t_{i+1}-2x) - \frac{(t_{i+1}-x)^2}{t_{i+1}-t_i}}{t_{i+2}-t_i}$$

Third order divided difference:

$$\begin{aligned}
&f[t_{i-1}, t_i, t_{i+1}, t_{i+2}] \\
&= \frac{f[t_i, t_{i+1}, t_{i+2}] - f[t_{i-1}, t_i, t_{i+1}]}{t_{i+2} - t_{i-1}} \\
&= \frac{1}{t_{i+2}-t_{i-1}} \left[\frac{(t_{i+2}+t_{i+1}-2x) - \frac{(t_{i+1}-x)^2}{t_{i+1}-t_i}}{t_{i+2}-t_i} - \frac{(t_{i+1}-x)^2}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} \right]
\end{aligned}$$

Now multiply by $(t_{i+2} - t_{i-1})$:

$$\begin{aligned}
&(t_{i+2}-t_{i-1})F(x) \\
&= \frac{(t_{i+2}+t_{i+1}-2x) - \frac{(t_{i+1}-x)^2}{t_{i+1}-t_i}}{t_{i+2}-t_i} - \frac{(t_{i+1}-x)^2}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)}
\end{aligned}$$

Let's work on the first term. Write it over common denominator $(t_{i+1} - t_i)$:

$$\begin{aligned}
&\frac{(t_{i+2}+t_{i+1}-2x) - \frac{(t_{i+1}-x)^2}{t_{i+1}-t_i}}{t_{i+2}-t_i} \\
&= \frac{\frac{(t_{i+2}+t_{i+1}-2x)(t_{i+1}-t_i) - (t_{i+1}-x)^2}{(t_{i+1}-t_i)}}{t_{i+2}-t_i} \\
&= \frac{(t_{i+2}+t_{i+1}-2x)(t_{i+1}-t_i) - (t_{i+1}-x)^2}{(t_{i+2}-t_i)(t_{i+1}-t_i)}
\end{aligned}$$

Now expand the numerator:

$$\begin{aligned}
&(t_{i+2}+t_{i+1}-2x)(t_{i+1}-t_i) - (t_{i+1}-x)^2 \\
&= (t_{i+2}+t_{i+1}-2x)(t_{i+1}-t_i) - (t_{i+1}^2 - 2t_{i+1}x + x^2) \\
&= t_{i+2}t_{i+1} - t_{i+2}t_i + t_{i+1}^2 - t_{i+1}t_i - 2xt_{i+1} + 2xt_i - t_{i+1}^2 + 2t_{i+1}x - x^2 \\
&= t_{i+2}t_{i+1} - t_{i+2}t_i - t_{i+1}t_i + 2xt_i - x^2 \\
&= (t_{i+2}-t_i)(t_{i+1}-t_i) + 2xt_i - x^2 - (t_{i+1}-t_i)^2
\end{aligned}$$

So the first term becomes:

$$\frac{(t_{i+2}-t_i)(t_{i+1}-t_i) + 2xt_i - x^2 - (t_{i+1}-t_i)^2}{(t_{i+2}-t_i)(t_{i+1}-t_i)} = 1 + \frac{2xt_i - x^2 - (t_{i+1}-t_i)^2}{(t_{i+2}-t_i)(t_{i+1}-t_i)}$$

Now the complete expression is:

$$\begin{aligned}
&(t_{i+2}-t_{i-1})F(x) \\
&= \left[1 + \frac{2xt_i - x^2 - (t_{i+1}-t_i)^2}{(t_{i+2}-t_i)(t_{i+1}-t_i)} \right] - \frac{(t_{i+1}-x)^2}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)}
\end{aligned}$$

Now combine over common denominator $(t_{i+2} - t_i)(t_{i+1} - t_{i-1})(t_{i+1} - t_i)$:

$$\begin{aligned}
&(t_{i+2} - t_{i-1})F(x) \\
&= \frac{(t_{i+2} - t_i)(t_{i+1} - t_{i-1})(t_{i+1} - t_i)}{(t_{i+2} - t_i)(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} \\
&\quad + \frac{[2xt_i - x^2 - (t_{i+1} - t_i)^2](t_{i+1} - t_{i-1})}{(t_{i+2} - t_i)(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} \\
&\quad - \frac{(t_{i+1} - x)^2(t_{i+2} - t_i)}{(t_{i+2} - t_i)(t_{i+1} - t_{i-1})(t_{i+1} - t_i)}
\end{aligned}$$

So the numerator becomes:

$$\begin{aligned}
N =& (t_{i+2} - t_i)(t_{i+1} - t_{i-1})(t_{i+1} - t_i) \\
&+ [2xt_i - x^2 - (t_{i+1} - t_i)^2](t_{i+1} - t_{i-1}) \\
&- (t_{i+1} - x)^2(t_{i+2} - t_i)
\end{aligned}$$

After expanding all terms and collecting like terms, we can factor the numerator as:

$$N = (t_{i+2} - t_i)(x - t_{i-1})(t_{i+1} - x) + (t_{i+1} - t_{i-1})(t_{i+2} - x)(x - t_i)$$

Therefore:

$$\begin{aligned}
&(t_{i+2} - t_{i-1})F(x) \\
&= \frac{(t_{i+2} - t_i)(x - t_{i-1})(t_{i+1} - x) + (t_{i+1} - t_{i-1})(t_{i+2} - x)(x - t_i)}{(t_{i+2} - t_i)(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} \\
&= \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} \cdot \frac{t_{i+1} - x}{t_{i+1} - t_i} + \frac{t_{i+2} - x}{t_{i+2} - t_i} \cdot \frac{x - t_i}{t_{i+1} - t_i}
\end{aligned}$$

This matches exactly the second piece of $B_i^2(x)$ from Example 3.25.

Case 4: $x \in [t_{i-1}, t_i)$ For $x \in [t_{i-1}, t_i)$:

- For $t = t_{i-1}$: $t \leq x$, so $(t - x)_+^2 = 0$
- For $t = t_i, t_{i+1}, t_{i+2}$: $t > x$, so $(t - x)_+^2 = (t - x)^2$

Compute the divided difference step by step:

$$\begin{aligned}
f[t_{i-1}] &= 0 \\
f[t_i] &= (t_i - x)^2 \\
f[t_{i+1}] &= (t_{i+1} - x)^2 \\
f[t_{i+2}] &= (t_{i+2} - x)^2
\end{aligned}$$

First order divided differences:

$$\begin{aligned}
f[t_{i-1}, t_i] &= \frac{(t_i - x)^2 - 0}{t_i - t_{i-1}} = \frac{(t_i - x)^2}{t_i - t_{i-1}} \\
f[t_i, t_{i+1}] &= \frac{(t_{i+1} - x)^2 - (t_i - x)^2}{t_{i+1} - t_i} \\
f[t_{i+1}, t_{i+2}] &= \frac{(t_{i+2} - x)^2 - (t_{i+1} - x)^2}{t_{i+2} - t_{i+1}}
\end{aligned}$$

Second order divided differences:

$$
\begin{aligned}
f[t_{i-1}, t_i, t_{i+1}] &= \frac{f[t_i, t_{i+1}] - f[t_{i-1}, t_i]}{t_{i+1} - t_{i-1}} \\
&= \frac{\frac{(t_{i+1}-x)^2-(t_i-x)^2}{t_{i+1}-t_i} - \frac{(t_i-x)^2}{t_i-t_{i-1}}}{t_{i+1} - t_{i-1}} \\
f[t_i, t_{i+1}, t_{i+2}] &= \frac{f[t_{i+1}, t_{i+2}] - f[t_i, t_{i+1}]}{t_{i+2} - t_i}
\end{aligned}
$$

Third order divided difference:

$$
\begin{aligned}
&f[t_{i-1}, t_i, t_{i+1}, t_{i+2}] \\
&= \frac{f[t_i, t_{i+1}, t_{i+2}] - f[t_{i-1}, t_i, t_{i+1}]}{t_{i+2} - t_{i-1}}
\end{aligned}
$$

Now let's compute each part systematically. First, simplify $f[t_i, t_{i+1}]$:

$$
\begin{aligned}
f[t_i, t_{i+1}] &= \frac{(t_{i+1} - x)^2 - (t_i - x)^2}{t_{i+1} - t_i} \\
&= \frac{(t_{i+1}^2 - 2t_{i+1}x + x^2) - (t_i^2 - 2t_i x + x^2)}{t_{i+1} - t_i} \\
&= \frac{t_{i+1}^2 - t_i^2 - 2x(t_{i+1} - t_i)}{t_{i+1} - t_i} \\
&= \frac{(t_{i+1} - t_i)(t_{i+1} + t_i) - 2x(t_{i+1} - t_i)}{t_{i+1} - t_i} \\
&= t_{i+1} + t_i - 2x
\end{aligned}
$$

Similarly, $f[t_{i+1}, t_{i+2}] = t_{i+2} + t_{i+1} - 2x$
Now compute $f[t_i, t_{i+1}, t_{i+2}]$:

$$
\begin{aligned}
f[t_i, t_{i+1}, t_{i+2}] &= \frac{f[t_{i+1}, t_{i+2}] - f[t_i, t_{i+1}]}{t_{i+2} - t_i} \\
&= \frac{(t_{i+2} + t_{i+1} - 2x) - (t_{i+1} + t_i - 2x)}{t_{i+2} - t_i} \\
&= \frac{t_{i+2} - t_i}{t_{i+2} - t_i} = 1
\end{aligned}
$$

Now compute $f[t_{i-1}, t_i, t_{i+1}]$:

$$
\begin{aligned}
f[t_{i-1}, t_i, t_{i+1}] &= \frac{f[t_i, t_{i+1}] - f[t_{i-1}, t_i]}{t_{i+1} - t_{i-1}} \\
&= \frac{(t_{i+1} + t_i - 2x) - \frac{(t_i-x)^2}{t_i-t_{i-1}}}{t_{i+1} - t_{i-1}}
\end{aligned}
$$

Now we have all components for the third order divided difference:

$$
\begin{aligned}
&f[t_{i-1}, t_i, t_{i+1}, t_{i+2}] \\
&= \frac{f[t_i, t_{i+1}, t_{i+2}] - f[t_{i-1}, t_i, t_{i+1}]}{t_{i+2} - t_{i-1}} \\
&= \frac{1 - \frac{(t_{i+1}+t_i-2x) - \frac{(t_i-x)^2}{t_i-t_{i-1}}}{t_{i+1}-t_{i-1}}}{t_{i+2} - t_{i-1}}
\end{aligned}
$$

Multiply by $(t_{i+2} - t_{i-1})$:

$$
\begin{aligned}
&(t_{i+2} - t_{i-1})F(x) \\
&= 1 - \frac{(t_{i+1} + t_i - 2x) - \frac{(t_i-x)^2}{t_i-t_{i-1}}}{t_{i+1} - t_{i-1}}
\end{aligned}
$$

Simplify the expression inside:

$$
\begin{aligned}
&(t_{i+2} - t_{i-1})F(x) \\
&= 1 - \frac{t_{i+1} + t_i - 2x}{t_{i+1} - t_{i-1}} + \frac{(t_i - x)^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}
\end{aligned}
$$

Now combine over common denominator $(t_{i+1} - t_{i-1})$:

$$
\begin{aligned}
&(t_{i+2} - t_{i-1})F(x) \\
&= \frac{t_{i+1} - t_{i-1} - (t_{i+1} + t_i - 2x)}{t_{i+1} - t_{i-1}} + \frac{(t_i - x)^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})} \\
&= \frac{t_{i+1} - t_{i-1} - t_{i+1} - t_i + 2x}{t_{i+1} - t_{i-1}} + \frac{(t_i - x)^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})} \\
&= \frac{-t_{i-1} - t_i + 2x}{t_{i+1} - t_{i-1}} + \frac{(t_i - x)^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}
\end{aligned}
$$

Now combine both terms over common denominator $(t_{i+1} - t_{i-1})(t_i - t_{i-1})$:

$$
\begin{aligned}
&(t_{i+2} - t_{i-1})F(x) \\
&= \frac{(-t_{i-1} - t_i + 2x)(t_i - t_{i-1}) + (t_i - x)^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}
\end{aligned}
$$

Expand the numerator:

$$
\begin{aligned}
&(-t_{i-1} - t_i + 2x)(t_i - t_{i-1}) + (t_i - x)^2 \\
&= (-t_{i-1} - t_i + 2x)(t_i - t_{i-1}) + (t_i^2 - 2t_i x + x^2) \\
&= -t_{i-1}t_i + t_{i-1}^2 - t_i^2 + t_i t_{i-1} + 2xt_i - 2xt_{i-1} + t_i^2 - 2t_i x + x^2 \\
&= t_{i-1}^2 - 2xt_{i-1} + x^2 \\
&= (x - t_{i-1})^2
\end{aligned}
$$

Notice that all terms with t_i cancel out.

Therefore:

$$
\begin{aligned}
&(t_{i+2} - t_{i-1})F(x) \\
&= \frac{(x - t_{i-1})^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}
\end{aligned}
$$

This matches exactly the first piece of $B_i^2(x)$ from Example 3.25.

Case 5: $x < t_{i-1}$ If $x < t_{i-1}$, then for all $t = t_{i-1}, t_i, t_{i+1}, t_{i+2}$, we have $t > x$, so $(t - x)_+^2 = (t - x)^2$. The fourth divided difference of a cubic polynomial is constant, and specifically:

$$
[t_{i-1}, t_i, t_{i+1}, t_{i+2}](t - x)^2 = 0
$$

since $(t - x)^2$ is a quadratic polynomial and the fourth divided difference of any polynomial of degree ≤ 2 is zero. Thus $F(x) = 0 = B_i^2(x)$.

Conclusion We have verified that for all $x \in \mathbb{R}$:

$$
(t_{i+2} - t_{i-1})[t_{i-1}, t_i, t_{i+1}, t_{i+2}](t - x)_+^2 = B_i^2(x)
$$

This completes the algebraic verification of Theorem 3.32 for the case $n = 2$.

Problem VII: Scaled Integral of B-splines

Problem Statement

Deduce from the Theorem on derivatives of B-splines that the scaled integral of a B-spline $B_i^n(x)$ over its support is independent of its index i, even if the spacing of the knots is not uniform.

Solution

The desired result is precisely Corollary 3.33. The scaled integral is defined as the average value of the B-spline over its support interval $[t_{i-1}, t_{i+n}]$, and it equals $1/(n+1)$.

The scaled integral on the left-hand side is:

$$\frac{1}{t_{i+n} - t_{i-1}} \int_{t_{i-1}}^{t_{i+n}} B_i^n(x)\, dx.$$

By Theorem 3.32 (the representation of a B-spline as a divided difference of the truncated power function), we have:

$$B_i^n(x) = [t_{i-1}, \ldots, t_{i+n}](t-x)_+^n.$$

Substituting this into the integral and exchanging the order of integration and the divided difference (which are both linear operations) yields:

$$\frac{1}{t_{i+n} - t_{i-1}} \int_{t_{i-1}}^{t_{i+n}} [t_{i-1}, \ldots, t_{i+n}](t-x)_+^n\, dx = [t_{i-1}, \ldots, t_{i+n}] \frac{1}{t_{i+n} - t_{i-1}} \int_{t_{i-1}}^{t_{i+n}} (t-x)_+^n\, dx.$$

Computing the integral of the truncated power function over its support:

$$\int_{t_{i-1}}^{t_{i+n}} (t-x)_+^n\, dx = \frac{(t-t_{i-1})^{n+1}}{n+1}.$$

Thus, the expression becomes:

$$[t_{i-1}, \ldots, t_{i+n}] \frac{(t-t_{i-1})^{n+1}}{n+1}.$$

Finally, applying Corollary 2.22 (which states that the n-th divided difference of t^{n+1} is 1, and by linearity, the n-th divided difference of $(t-t_{i-1})^{n+1}$ is also 1), we obtain:

$$\frac{1}{n+1}[t_{i-1}, \ldots, t_{i+n}](t-t_{i-1})^{n+1} = \frac{1}{n+1}.$$

This completes the proof. The scaled integral is indeed independent of the index i and the knot spacing, depending only on the degree n.

Problem VIII: Symmetric Polynomials

Problem Statement

We have a theorem on expressing complete symmetric polynomials as divided difference of monomials.

(a) Verify this theorem for $m = 4$ and $n = 2$ by working out the table of divided difference and comparing the result to the definition of complete symmetric polynomials.

(b) Prove this theorem by the lemma on the recursive relation on complete symmetric polynomials.

Solution

Theorem 3.46 (Complete symmetric polynomials as divided difference of monomials)

The complete symmetric polynomial of degree $m-n$ in $n+1$ variables is the nth divided difference of the monomial x^m, i.e.

$$\forall m \in \mathbb{N}^+, \forall i \in \mathbb{N}, \forall n = 0, 1, \ldots, m,$$

$$\tau_{m-n}(x_i, \ldots, x_{i+n}) = [x_i, \ldots, x_{i+n}]x^m. \tag{3.56}$$

(a) **Verification for $m = 4$ and $n = 2$**

Let $t_1 = x_i, t_2 = x_{i+1}, t_3 = x_{i+2}$. We compute $[t_1, t_2, t_3]x^4$ using the divided difference table:

$$\begin{array}{c|ccc}
x & Q_{[]} & Q_{[]} & Q_{[]} \\
\hline
t_1 & t_1^4 & & \\
 & & \dfrac{t_2^4 - t_1^4}{t_2 - t_1} & \\
 & & & \dfrac{\dfrac{t_3^4 - t_2^4}{t_3 - t_2} - \dfrac{t_2^4 - t_1^4}{t_2 - t_1}}{t_3 - t_1} \\
t_2 & t_2^4 & & \\
 & & \dfrac{t_3^4 - t_2^4}{t_3 - t_2} & \\
t_3 & t_3^4 & &
\end{array}$$

First, compute the first order divided differences:

$$[t_1, t_2]x^4 = \frac{t_2^4 - t_1^4}{t_2 - t_1} = t_1^3 + t_1^2 t_2 + t_1 t_2^2 + t_2^3$$

$$[t_2, t_3]x^4 = \frac{t_3^4 - t_2^4}{t_3 - t_2} = t_2^3 + t_2^2 t_3 + t_2 t_3^2 + t_3^3$$

Now compute the second order divided difference:

$$\begin{aligned}
[t_1, t_2, t_3]x^4 &= \frac{[t_2, t_3]x^4 - [t_1, t_2]x^4}{t_3 - t_1} \\
&= \frac{(t_2^3 + t_2^2 t_3 + t_2 t_3^2 + t_3^3) - (t_1^3 + t_1^2 t_2 + t_1 t_2^2 + t_2^3)}{t_3 - t_1} \\
&= \frac{t_2^2 t_3 + t_2 t_3^2 + t_3^3 - t_1^3 - t_1^2 t_2 - t_1 t_2^2}{t_3 - t_1}
\end{aligned}$$

Factor numerator by recognizing it as $\tau_2(t_1, t_2, t_3)$:

$$\tau_2(t_1, t_2, t_3) = t_1^2 + t_1 t_2 + t_1 t_3 + t_2^2 + t_2 t_3 + t_3^2$$

Indeed, one can verify that:

$$[t_1, t_2, t_3]x^4 = t_1^2 + t_1 t_2 + t_1 t_3 + t_2^2 + t_2 t_3 + t_3^2 = \tau_2(t_1, t_2, t_3)$$

Since $m = 4$, $n = 2$, we have $m - n = 2$, which matches the degree of τ_2. This verifies the theorem for this case.

(b) **Proof using recursive relation**

The complete symmetric polynomials satisfy the recurrence:

$$\tau_k(x_i, \ldots, x_{i+n}) = x_i \cdot \tau_{k-1}(x_i, \ldots, x_{i+n}) + \tau_k(x_{i+1}, \ldots, x_{i+n})$$

with initial conditions $\tau_0(x_i, \ldots, x_{i+n}) = 1$ and $\tau_k(\emptyset) = 0$ for $k > 0$.

We prove by induction on n that:

$$\tau_{m-n}(x_i, \ldots, x_{i+n}) = [x_i, \ldots, x_{i+n}]x^m$$

Base case: $n = 0$

$$[x_i]x^m = x_i^m = \tau_m(x_i)$$

This matches since $m - 0 = m$.

Inductive step: Assume the theorem holds for $n - 1$. For n, we use the divided difference recurrence:

$$[x_i, \ldots, x_{i+n}]x^m = \frac{[x_{i+1}, \ldots, x_{i+n}]x^m - [x_i, \ldots, x_{i+n-1}]x^m}{x_{i+n} - x_i}$$

By induction hypothesis:

$$[x_{i+1}, \ldots, x_{i+n}]x^m = \tau_{m-(n-1)}(x_{i+1}, \ldots, x_{i+n}) = \tau_{m-n+1}(x_{i+1}, \ldots, x_{i+n})$$
$$[x_i, \ldots, x_{i+n-1}]x^m = \tau_{m-(n-1)}(x_i, \ldots, x_{i+n-1}) = \tau_{m-n+1}(x_i, \ldots, x_{i+n-1})$$

Now using the recurrence relation for complete symmetric polynomials:

$$\tau_{m-n+1}(x_i, \ldots, x_{i+n-1}) = x_i \cdot \tau_{m-n}(x_i, \ldots, x_{i+n-1}) + \tau_{m-n+1}(x_{i+1}, \ldots, x_{i+n-1})$$

and

$$\tau_{m-n+1}(x_{i+1}, \ldots, x_{i+n}) = x_{i+n} \cdot \tau_{m-n}(x_{i+1}, \ldots, x_{i+n}) + \tau_{m-n+1}(x_{i+1}, \ldots, x_{i+n-1})$$

Substituting these into the divided difference expression and simplifying, we obtain:

$$[x_i, \ldots, x_{i+n}]x^m = \tau_{m-n}(x_i, \ldots, x_{i+n})$$

This completes the inductive proof.